Matrix Derivation

$\nabla tr(X) = I \quad \frac{\partial}{\partial a_{ij}}(a_{11} + \cdots + a_{NN}) = \delta_{ij}$

$\nabla tr(AX) = A$

$\nabla tr(AX^{-1}) = -X^{-H}AX^{-H}$

$(X + dX)^{-1} = (I + X^{-1}dX)^{-1}X^{-1} = X^{-1} - X^{-1}dXX^{-1}$

$dtr(AX^{-1}) = -tr(AX^{-1}dXX^{-1})$

$\frac{\partial}{\partial x} x^H A x = 2Ax$

$\frac{\partial}{\partial A} x^H A x = xx^H$

$\frac{\partial}{\partial A} x^H A^{-1} x =$

e.g. SPICE $f = tr(\hat{R}^{-1}R) + tr(\hat{R}R^{-1})$

$\underset{M\times M}{\nabla_R f} = \hat{R}^{-1} - R^{-H}\hat{R}R^{-H}$, $R = APA^H + \sigma^2 I = \sum_{n=1}^{N} p_n a_n a_n^H + \sigma^2 I$

$\frac{\partial}{\partial p_n} f = \langle \nabla_R f, \frac{\partial}{\partial p_n} R \rangle = tr\{\underset{M\times 1}{a_n} a_n^H(\hat{R}^{-1} - R^{-1}\hat{R}R^{-1})\} = a_n^H(\hat{R}^{-1} - R^{-1}\hat{R}R^{-1})a_n$

KKT : $A^H R^{-1}\hat{R}R^{-1}A = A^H\hat{R}^{-1}A$

$\nabla \|X\|_F^2 = \nabla tr(X^H X) = 2X$

$dtr(X^H X) = tr(X^H dX) + tr(dX^H X) + tr(dX^H dX)$

$= \langle 2X, dX \rangle + O(dX)$

$\nabla |X| = |X| \cdot X^{-H} \quad d|X| = |X + dX| - |X| = |X|\, tr(X^{-1}dX)$

$\nabla \log|X| = X^{-H}$

$\otimes$, $\odot$, vec

Kronecker Product $A \otimes B = \begin{bmatrix} a_{11}B & a_{12}B \\ a_{21}B & a_{22}B \end{bmatrix}_{4\times 4}$

Khatri-Rao Product $\underset{M_1\times N}{A} \odot \underset{M_2\times N}{B} = (\underset{(M_1\times M_2)\times N}{a_1 \otimes b_1 \ \cdots \ a_N \otimes b_N})$

e.g. $A \odot B = \begin{bmatrix} a_{11}\vec{b}_{1\,2\times 1} & a_{21}\vec{b}_2 \\ a_{12}\vec{b}_1 & a_{22}\vec{b}_2 \end{bmatrix}_{4\times 2}$ $A \odot B$ 是 $A \otimes B$ 的第一、四列

$vec(AXB) = (B^T \otimes A)\, vec(X) = (B^T \odot A)\, diag(X)$,

if X is diagonal

e.g. 2D-DFT $Y = FXF$, $vec(Y) = (F^T \otimes F)\, vec(X)$

Covariance $R = APA^H + \sigma^2 I$, $vec(R) = (A^* \otimes A)\, vec(P) + \sigma^2 vec(I)$

$= (A^* \odot A)\, p + \sigma^2 vec(I)$

$tr(AB) = vec(A^T)^T vec(B) = vec(A^H)^H vec(B)$

$tr(ABCD) = vec(A^T)^T (D^T \otimes B)\, vec(C)$